

ROBERT ADRIEL MOSTOGHIU PAUN

Data Scientist

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EXECUTIVE SUMMARY

I build large-scale predictive models and automated ETL pipelines, with 8+ years of experience working across terabyte-scale datasets in distributed HPC environments. Currently at the ARC Dark Matter Centre of Excellence, I lead international projects optimising high-fidelity simulations, achieving a **40% accuracy increase** in HPC environments and a **98% reduction in engineering cycle times** through AI orchestration. My core stack is Python and C/C++, with hands-on experience in scikit-learn and PyTorch across statistical inference and large-scale numerical analysis.

WORK EXPERIENCE

Postdoctoral Research Associate

ARC Centre for Dark Matter and Particle Physics, Swinburne University

February 2022 – Present

Melbourne, VIC, Australia

- **Model Development & Validation:** Engineered custom extensions for the GIZMO computational framework (GIZMO-PBHEF) to model complex system dynamics in distributed High Performance Computing (HPC) environments. Developed and validated novel simulation techniques that achieved a **40% improvement in model accuracy** by identifying and correcting systemic errors within terabyte-scale predictive models.
- **Data Pipeline Engineering:** Architected **end-to-end Extract, Transform, Load (ETL) workflows** in Python and SQL to transform terabyte-scale raw simulation data into structured, high-fidelity formats for distributed HPC processing.
- **Anomaly Detection:** Applied A/B testing, feature engineering, statistical thresholds, and population-level diagnostics to identify anomalies and **distinguish genuine signals from computational noise**.
- **AI Orchestration:** Built the "SAGE Tree Converter", an LLM-powered CLI framework using an adapter design pattern and persistent knowledge base to automate complex hierarchical data conversion, cutting a 1-3 month manual engineering cycle to a single day (**98% reduction in cycle time**).
- **Platform Administration:** Built and administered a custom collaboration platform for a **100-member international research consortium** (The Three Hundred), replacing a commercial Slack setup and **avoiding ongoing licensing costs**.
- **Governance & Operations:** Appointed to the Swinburne Time Allocation Committee (STACK) to **assess and rank competing project proposals** against technical and scientific merit criteria to allocate high-value compute resources across an international research consortium. Managed compliance and operational infrastructure for cross-institutional research partnerships.
- **Stakeholder Engagement & Leadership:** Supervised **3 undergraduate** and **2 postgraduate** researchers. Delivered weekly progress reports on technical challenges and outputs to a cross-disciplinary team spanning astrophysics, experimental physics, and engineering.

PhD Researcher

Universidad Autónoma de Madrid

September 2016 – November 2020

Madrid, Spain

- **Data Integration & Standardisation:** Developed Python-based workflows (NumPy/Pandas) to integrate and standardise **terabyte-scale datasets** distributed across multiple international HPC facilities, enabling large-scale comparative analysis for the "The Three Hundred" consortium.
- **Statistical Benchmarking:** Ran rigorous statistical benchmarking of multiple modelling architectures to isolate numerical artefacts from physical signals, establishing **reproducible evaluation criteria** across model comparisons.
- **Technical Communication:** Translated complex model outputs for both technical and non-technical audiences across **6 international conferences** and **9 peer-reviewed publications**.

EDUCATION

Universidad Autónoma de Madrid

Madrid, Spain

Doctor of Philosophy in Computational Astrophysics

September 2016 – November 2020

- **Awards:** Graduated with Summa Cum Laude (2020). Awarded a Doctorate Honourable Mention (2022).

Universidad Autónoma de Madrid

Madrid, Spain

Master of Science in Cosmology and Particle Physics

September 2015 – June 2016

- **Awards:** General enrolment funding and academic stipend by the Spanish Ministry of Education

Universidad Autónoma de Madrid

Madrid, Spain

Bachelor's Degree in Physics

September 2011 – June 2015

- **Awards:** General enrolment funding and academic stipend by the Spanish Ministry of Education

PROFESSIONAL DEVELOPMENT

Commercialisation Training Program for Medical Researchers

Melbourne, Australia

AAMRI VIC

September 2025 – December 2025

- Selected for a 20-person cohort to develop commercialisation pathways for cross-sector digital health projects. Completed the program with an investor pitch night, presenting technical research as a commercial proposition to an industry panel.

Research Translation & Design Thinking

Geneva, Switzerland

IdeaSquare, CERN

July 2025

- Completed an intensive program in research translation and design thinking, covering commercial literacy, visual storytelling, and communicating technical work to non-specialist audiences.

SKILLS

Programming & Databases: Python (NumPy, SciPy, Pandas, Polars, Seaborn, Matplotlib), Jupyter, SQL (PostgreSQL, MySQL), R, C/C++.

Data Science & Modelling: Numerical & Predictive Modelling, Machine Learning (scikit-learn, PyTorch), Generative AI, AI Orchestration (CLIs), Bayesian Inference, CI/CD, Large-Scale Data Analysis, Data Visualisation

Developer Tools: High-Performance Computing (HPC), Cloud Services (Azure Databricks), Dashboards (Power BI, Excel), Git, Docker, Linux Shell Script, SLURM, LaTeX

Languages: English (fluent), Spanish (native), Romanian (native)